

Equity Pesa Link P2P API Design Specification Document

Version Draft

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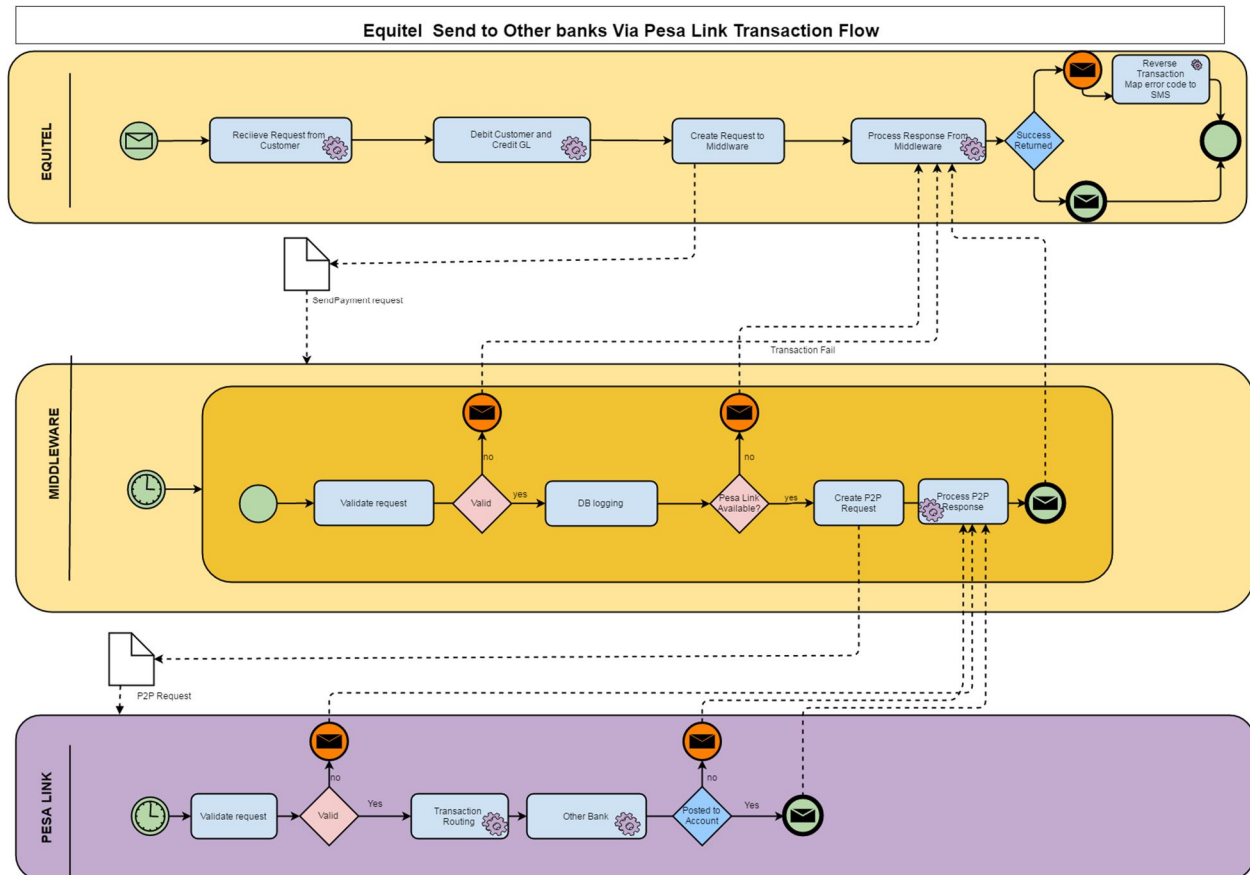
1.0 Introduction

This document defines the API interface linking Equitel and Pesa Link P2P service. The API is exposed as XML over HTTP. The API provides herein below functions:

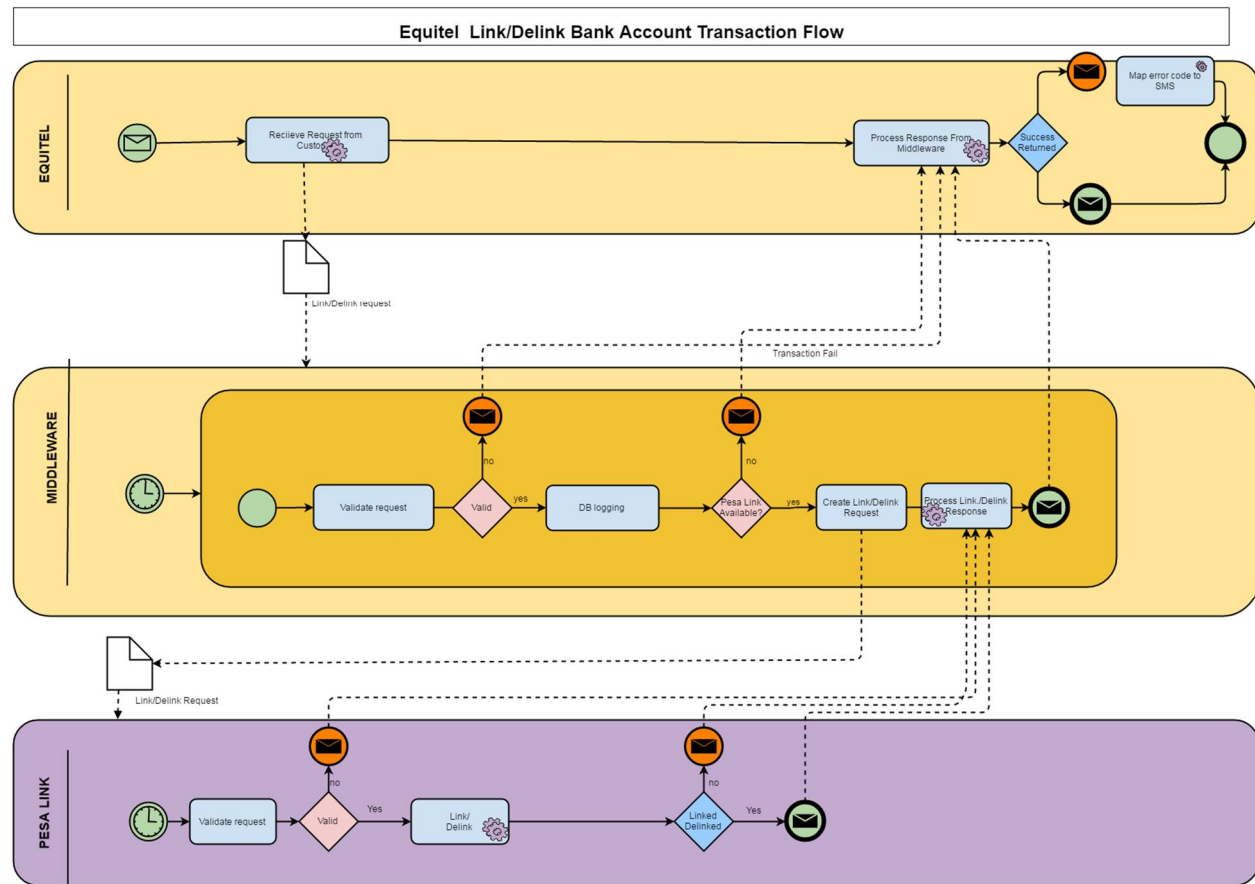
- Mobile Number Inquiry: - used by the sender to confirm recipient linked Bank accounts.
- Send Money: - (P2P) Used to send money to Other Banks.
- Link Account to Mobile :- Used by receiver to link Bank Account to Mobile Number
- Delink Account from Mobile :- Used by receiver to delink Bank Account to Mobile Number

1.1 Transaction Flows

1.1.1 Send Money P2P:



1.1.2 Link/Delink:



1.2 API Description

The API is XML on HTTP. The URL will be shared separately.

Kba transactions url=<http://10.14.2.7:7001/kbapayhandler/>
Kba register.url= [http:// 10.14.2.7:7001/kbaRegister/](http://10.14.2.7:7001/kbaRegister/)
kba inquire url= [http:// 10.14.2.7:7001/kbaInquire/](http://10.14.2.7:7001/kbaInquire/)

1.2.1 Inquire API Fields

Source Service Tag	Data Type	Length	Variable Length	Mandatory /Optional	Description Default/Expected Values
mobileNumber	String/text	50	Y	M	Mobile Number
signature	String/text	200	Y	M	Signature This field is a checksum. Equitel will sign this field with SSL private Key. The following fields are first concatenated.

					(messageld + mobileNumber + channel + amount+ accountType) in that order (Salt to be shared) The resulting text is then signed with Private Key and Base64 encoded.
messageld	String/text	35	Y	M	Message Id
channel	String/text	35	Y	M	channel

1.2.2 Send Money API Fields

Source Service Tag	Data Type	Length	Variable Length	Mandatory /Optional	Description Default/Expected Values
amount	String/text	13	Y	M	Transaction Amount
sender	String/text	35	Y	M	Sender Name
senderAccount	String/text	20	Y	M	Sender Bank Account
destPhone	String/text	15	Y	M	Beneficiary Mobile Number
destAccount	String/text	20	Y	M	Beneficiary Bank Account
destBankCode	String/text	5	Y	M	Beneficiary Bank Code
currency	String/text	3	N	M	Default to KES
signature	String/text	200	Y	M	This field is a checksum. Equitel will sign this field with SSL private Key. The following fields are first concatenated. (SALT + destAccount + destBankCode + rrn + amount+ accountType) in that order (Salt to be shared) The resulting text is then signed with Private Key and Base64 encoded.

destName	String/text	35	Y	M	Beneficiary Name
accountType	String/text	3	N	M	Valid Values: A2A or A2M
rrn	String/text	12	N	M	Transaction reference Number
channel	String/text	50	Y	M	Valid value: MFS
isInCharge	String/text	2	Y	O	If Charging is configured at the source the value should be Y
charge	String/text	5	Y	C	If <i>isInCharge</i> is Y the value to be charged should be supplied here

1.2.1 Link Bank Account API Fields

Source Service Tag	Data Type	Length	Variable Length	Mandatory /Optional	Description Default/Expected Values
accountNumber	String/text	13	N	M	Account to Be linked
nationalId	String/text	50	Y	O	National Id
firstName	String/text	50	Y	M	Customer Name/Account Name
mobileNumber	String/text	12	N	M	Customer Mobile Number to be Linked
signature	String/text	200	Y	M	This field is a checksum. Equitel will sign this field with SSL private Key. The following fields are first concatenated. (SALT + accountNumber + mobileNumber + messageld) in that order (Salt to be shared) The resulting text is then signed with Private Key and Base64 encoded.
timestamp	String/text	50	Y	O	Date time

channel	String/text	20	Y	M	Valid value: MFS
actionType	String/text	20	Y	M	Valid Values LINK, DELINK

1.3 Sample requests

1.3.1 Send to Other Bank Payment Request

```
<com.equitybank.payment.interbank.models.InterbankReq >
  <messageId>23243437</messageId>
  <amount>500</amount>
  <senderName>Geoff</senderName>
  <senderAccount>1111</senderAccount>
  <destPhone>343434</destPhone>
  <destAccount>4452323</destAccount>
  <destBankCode>01</destBankCode>
  <currency>KES</currency>
  <destName>Wam</destName>
  <rrn>2324343</rrn>
  <accountType>A2A</accountType>
  <channel>MFS</channel>
  <isInCharge>Y</isInCharge>
  <charge>200</charge>
  <signature>ZAYqMffMrUDFcArlo6nKkWM6PEkTHkLEKM3Ehf82PQgGLqEY24oBZQVYMpBoqMYJaAJfJDckl7uS7wQ2uhcgBNhvyo4lml5ildyX/t0io4OZ
guZMA5fGowFJPnIz3ZoHqHe2NUL/iAGtMm6GulhnK9xvhfk+V+FunAj56FRZOhZvUgc+ooMEqGM93PpyKPG0m3Y84kvH9qZ/W40sWIUevnAFIm9
Vx2MPPTWtafPkLEpWmjfMi4pAG+56ad7oWOSfojP1X9VbsUqCRYnemoK1AzfrENCx2yFwr2mWm28glkBujH1JHVdB83iqFcDv9b14EplgfPh4okMao
FSZHkmA==</signature>

</com.equitybank.payment.interbank.models.InterbankReq >
```

1.3.2 Send to Other Bank Payment response

```
<com.equitybank.payment.interbank.models.InterbankRes>
  <messageId>23243437</messageId>

  <rrn>2324343</rrn>

  <responseCode>0</responseCode>

  <responseDescription>Paid Success</responseDescription>

</com.equitybank.payment.interbank.models.InterbankRes>
```

1.3.3 Link Account:

```
<com.equitybank.payment.interbank.models.RegisterCustomerReq>
  <customer>
    <firstName>John</firstName>
    <lastName>Doe</lastName>
    <accountNumber>1233366</accountNumber>
    <mobileNumber>2543556333</mobileNumber>
    <emailAddress>johndoe@mail.com</emailAddress>
  </customer>
  <messageId>2324343</messageId>
  <timestamp>2016-04-14</timestamp>
  <signature>xxx</signature>
  <channel>MFS</channel>
  <actionType>LINK</actionType>
</com.equitybank.payment.interbank.models.RegisterCustomerReq>
```

1.3.4 Link Account Response:

```
<com.equitybank.payment.interbank.models.RegisterCustomerRes>
  <messageId>2324343</messageId>
  <actionType>LINK</actionType>
  <responseCode>0</responseCode>
  <responseDescription>Linked Successful</responseDescription>
</com.equitybank.payment.interbank.models.RegisterCustomerRes>
```

1.3.5 Inquire Account Request:

```
<com.equitybank.payment.interbank.models.KbaInquireReq>
  <customer>
    <mobileNumber>254763388149</mobileNumber>
  </customer>
  <messageId>23243436</messageId>
  <timestamp>2016-04-14</timestamp>
  <signature>xxxsdfsdfsdfs</signature>
  <channel>MFS</channel>
</com.equitybank.payment.interbank.models.KbaInquireReq>
```

1.3.6 Inquire Account Response:

```
<com.equitybank.payment.interbank.models.KbaInquireRes>

  <messageId>23243437</messageId>

  <responseCode>0</responseCode>

  <bankList>

    <entry>

      <string>01</string>

      <string>KCB Joshua</string>

    </entry>

    <entry>

      <string>70</string>

      <string>Family Joshua</string>

    </entry>

  </bankList>

  <entry>
```

```

<string>32</string>

<string>StanBic Joshua</string>

</entry>

</bankList>

</com.equitybank.payment.interbank.models.KbInquireRes>

```

1.4 Error Codes

Error Code	Description Default/Expected Values
-1	Validation Missing fields
-3	Unable to process request
-6	Checksum Error
-10	Pesa Link unavailable
107	refer to card issuer
110	invalid amount
114	Account Not Found
119	All the other error codes 101 Decline Decline, expired card 102 Decline Decline, suspected fraud 103 Decline Decline, card acceptor contact acquirer 104 Decline Decline, restricted card 105 Decline Decline, card acceptor call acquirer's security department 106 Decline Decline, allowable PIN tries exceeded 108 Decline Decline, refer to card issuer's special conditions 109 Decline Decline, invalid merchant 111 Decline Decline, invalid card number 112 Decline Decline, PIN data required 113 Decline Decline, unacceptable fee requested

	115 Decline Decline, requested function not supported
	116 Decline, no funds Decline, not sufficient funds
	117 Decline Decline, incorrect PIN
	118 Decline Decline, no card record
	119 Decline Decline, transaction not permitted to cardholder
	120 Decline Decline, transaction not permitted to terminal
	121 Decline Decline, exceeds withdrawal amount limit
	122 Decline Decline, security violation
	123 Decline Decline, exceeds withdrawal frequency limit
	124 Decline Decline, violation of law
	125 Decline Decline, card not effective
	126 Decline Decline, invalid PIN block
	127 Decline Decline, PIN length error
	128 Decline Decline, PIN key synch error
	129 Decline Decline, suspected counterfeit card